WORKED SOLUTION

STEP I 1999, Question 13

Probability and Statistics • Indicator Variables

STEP I 1999, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s1-q13

The key idea

Don't count the units directly. Count the *gaps* instead: the number of units is one more than the number of places where neighbouring magnets stand apart. Whether a gap opens depends only on whether the two neighbours point the same way, and these $N-1$ events turn out to be independent fair coin tosses. So the number of gaps is binomial, and both the expectation and the variance follow at once.

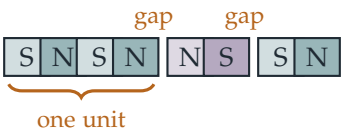
Solution

Setting up the model

We take "placed randomly" to mean that each magnet is equally likely to point either way, independently of the others. Call a magnet *type A* if its north pole is on the right (SN) and *type B* if its north pole is on the left (NS). Let $X_1, \ldots, X_N$ be the types, read from left to right.

Look at two neighbours. The ends facing each other are the right end of the first magnet and the left end of the second.

- Same type (A then A): the facing ends are N and S, which are opposite, so they **join**. (B then B is the same with S and N.)
- Different types (A then B): the facing ends are N and N, so they **stand apart**. (B then A gives S and S.)



So a gap appears between magnets i and $i+1$ exactly when $X_i \neq X_{i+1}$. Define, for

$i = 1, \ldots, N-1$,

$$G_i = \begin{cases} 1 & \text{if } X_i \neq X_{i+1} \text{ (a gap)}, \\ 0 & \text{if } X_i = X_{i+1} \text{ (joined)}. \end{cases}$$

Each gap splits the line into one more piece, so the number of separate units is

$$U = 1 + G_1 + G_2 + \cdots + G_{N-1}.$$

The distribution of the gaps

Each G_i is 1 with probability $\frac{1}{2}$: whatever X_i is, X_{i+1} differs from it with probability $\frac{1}{2}$.

Ask yourself The events "gap at i" and "gap at $i+1$" both involve magnet $i+1$. Are the G_i really independent?

They are, even though neighbouring G's share a magnet. Fix the type of the first magnet, X_1. Then the sequence $G_1, \ldots, G_{N-1}$ determines $X_2, X_3, \ldots, X_N$ one at a time (X_{i+1} equals X_i if $G_i = 0$ and is the other type if $G_i = 1$), and conversely the types determine the G's. So, for either value of X_1, the 2^{N-1} equally likely choices of $(X_2, \ldots, X_N)$ correspond one-to-one with the 2^{N-1} possible sequences $(G_1, \ldots, G_{N-1})$. Every sequence of gaps therefore has probability $1/2^{N-1}$, which says exactly that the G_i are independent, each 0 or 1 with probability $\frac{1}{2}$.

Hence $G_1 + \cdots + G_{N-1} \sim \mathrm{B}\left(N-1, \frac{1}{2}\right)$.

Expectation and variance

A $\mathrm{B}(n, p)$ variable has mean np and variance $np(1-p)$. With $n = N-1$, $p = \frac{1}{2}$, and remembering that adding the constant 1 shifts the mean but not the variance,

$$\mathrm{E}(U) = 1 + \tfrac{1}{2}(N-1) = \frac{N+1}{2},$$
$$\mathrm{Var}(U) = (N-1) \cdot \tfrac{1}{2} \cdot \tfrac{1}{2} = \frac{N-1}{4}.$$

If you prefer not to quote the binomial results: $\mathrm{E}(G_i) = \frac{1}{2}$ and, since $G_i^2 = G_i$, $\mathrm{Var}(G_i) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. Linearity gives the mean, and independence lets us add the $N-1$ variances.

Checks

- $N = 1$: one magnet is always one unit, and the formulae give mean 1 and variance 0.
- $N = 2$: the two magnets form 1 or 2 units with probability $\frac{1}{2}$ each, so the mean is $\frac{3}{2}$ and the variance is $\frac{5}{2} - \frac{9}{4} = \frac{1}{4}$, as predicted.
- An exact enumeration over all 2^N arrangements for $N \leq 12$ reproduced $\frac{N+1}{2}$ and $\frac{N-1}{4}$ exactly, and a simulation with $N = 20$ gave a mean of about 10.5 and a variance of about 4.73 (predicted 10.5 and 4.75).

Common slips

- Thinking that two magnets of the *same* orientation repel. They face each other with opposite poles (N to S), so they attract and join.
- Adding variances without justifying independence. Here the independence is true but not obvious, because neighbouring gaps share a magnet.
- Writing $\mathrm{Var}(U) = \frac{N}{4}$ by counting N gaps instead of $N-1$, or forgetting the extra 1 in the mean.

Beyond STEP

Counting units by counting the boundaries between them is the method of *indicator variables*, one of the most useful tools in probability: write a count as a sum of 0–1 variables, then use linearity of expectation, which needs no independence at all. Here U counts the *runs* in a random sequence of two symbols. Run counts are the basis of the Wald–Wolfowitz runs test in statistics, which checks whether a sequence looks random. For more on indicators and their variances, see Blitzstein and Hwang, *Introduction to Probability*.

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